

# Arjun Dahal

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## Education

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### PhD in Computer Science

Teaching Assistant Scholarship

University of Texas at Arlington

Aug 2022 – May 2027

### Bachelor's in Electronics and Communication Engineering

Ranked in top 100 of 12,000 entrants, Full Scholarship

IOE, Tribhuvan University

Oct 2018

**Thesis:** Nepali Speech Recognition using RNN-CTC Model

## Experience

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### Graduate Teaching Assistant

University of Texas at Arlington

Arlington, TX

Aug 2022 – Present

- Conduct research on generating natural test cases for fairness testing and mitigating bias for ML models.
- Assist in teaching courses related to Programming, Software Development, and Software Testing.

### Solutions Engineer

Guardsix (Formerly Logpoint)

Lalitpur, Nepal

2018 – 2022

- Solved software, system, and network issues in the production environment.
- Led the deployment and monitoring of SIEM/SOAR solutions across cloud and on-prem environments, developed custom analytics dashboards, alert pipelines, and event-driven orchestration scripts.
- Mentored new hires and led biweekly interdepartmental knowledge transfer sessions.

## Research Projects

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### Bias Mitigation for Pretrained Classifiers

2025 - Present

- Developed a diagnostic-guided bias mitigation approach for pretrained tabular classifiers that applies minimal, dataset-adaptive changes to reduce the use of sensitive attribute information during the decision-making process.

### Generation of Natural Test Cases for Fairness Testing

2022 - 2025

- Implemented a two-stage approach that either simply flips the sensitive attribute or uses an adversarial autoencoder for precise changes in related attributes for tabular instances. The generated instances are natural and high-proximity counterfactuals outperforming the state-of-the-art baselines in 8 out of 11 cases across 7 datasets.
- Applied Combinatorial Testing in the latent space of a Variational Autoencoder to efficiently generate natural test cases (improvement of 43% in naturalness over the baseline) for testing the fairness of black box models.

### Nepali Speech Recognition

2017 - 2018

- Applied a Recurrent Neural Network along with Connectionist Temporal Classification (CTC) loss for end-to-end Nepali language recognition.

## Publications

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**Dahal, A.**, Lei, Y., Kacker, R. N., & Kuhn, D. R. (2026). TabChange: An Approach to Natural Changes in Tabular Data. Under Review.

**Dahal, A.**, Shree, S., Lei, Y., Kacker, R. N., & Kuhn, D. R. (2025). Fairness Testing of Machine Learning Models Using Combinatorial Testing in Latent Space. *In Proceedings of the 2025 IEEE International Conference on Software Testing, Verification and Validation Workshops (ICSTW)*. IEEE.

Regmi, P., **Dahal, A.**, & Joshi, B. (2019). Nepali Speech Recognition Using RNN-CTC Model. *International Journal of Computer Applications*, 178(31), 1–6.

## Skills

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**Programming:** Python, C; Frameworks: Deep Learning (PyTorch), Machine Learning (Scikit-Learn), NumPy, Matplotlib; Domains: Tabular, NLP

**Troubleshooting:** System, Network, and Software debugging and troubleshooting

**Leadership:** IT Coordinator at Nepalese Student Association, University of Texas at Arlington (NSA-UTA)